

Run 02 | Business exit

India

IV only

Target Entrepreneurial/business exit in the past 12 months; binary 1 vs. 0
Sample 27,106 respondents **Coverage** 1 country; 2015–2022
Positive cases 1,402 (5.17%) **Case time** 50s
Train / val. / test 18,434 / 3,252 / 5,420 (68% / 12% / 20%, stratified)
SCT predictors (12): Self-efficacy, Opportunity perception, Fear of failure, Knowing entrepreneurs, Entrepreneurial intention, Media coverage, Business angel, Established business owner, Good career choice, Ease of starting a business, High status, Social enterprise.

Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 25.

Controls (0): None. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: H2O GBM (0.8243). Highest F1: CatBoost (0.3426). The all-negative accuracy baseline is 94.83% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
H2O GBM	0.8243	0.3605	94.32	42.31	27.50	0.3333	5s
XGBoost	0.8232	0.3579	94.48	44.10	25.36	0.3220	<1s
CatBoost	0.8220	0.3553	92.99	33.22	35.36	0.3426	10s
Random Forest	0.8164	0.3559	92.99	32.88	34.29	0.3357	1s
Lasso	0.8109	0.2548	92.53	28.22	28.93	0.2857	8s
Elastic Net	0.8105	0.2605	94.15	38.65	22.50	0.2844	8s
Ridge	0.8097	0.2644	94.23	40.12	23.93	0.2998	10s
Logistic Regression	0.7994	0.2910	94.24	40.70	25.00	0.3097	<1s
Naive Bayes	0.7595	0.1616	92.86	28.16	24.64	0.2629	6s
Decision Tree	0.6048	0.2011	94.06	37.50	22.50	0.2812	<1s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Business angel	147.52	26.25
2	Entrepreneurial intention	87.22	15.52
3	Knowing entrepreneurs	25.03	4.45
4	Fear of failure	19.06	3.39
5	Opportunity perception	17.06	3.04
6	Good career choice	16.24	2.89
7	Business angel (missing indicator)	14.27	2.54
8	Ease of starting a business	14.07	2.50

Share is the percentage of total forest impurity importance across all 49 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Investment environment	11.31	2.01
2	Enterprise conditions	8.41	1.50
3	Social capital	8.37	1.49

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 5 batch, 29 September 2026; 02_obj5_india_iv_EXIT_ENT_harmonized. Scores and importance are taken directly from saved CSV outputs.